Team Work Distribution for AI-Powered HR Assistant

1 Introduction

To ensure efficient development, minimal blocking, and parallel execution, the project is divided among five developers with clearly defined roles and responsibilities.

2 Team Distribution

2.1 Developer 1 — AI / Orchestrator Engineer

Responsibilities:

- LLM integration and configuration
- Tool definition and tool-calling logic
- Orchestrator and agent loop implementation
- Prompt engineering
- Response validation and sanitization

Main Tasks:

- Implement all tools (salary, leave, HR policy search)
- Build the orchestrator class
- Implement agent loop (LLM $\rightarrow$ tools $\rightarrow$ results)
- Create LLM provider abstraction
- Validate and sanitize outputs

Dependencies:

- Requires: Database (Dev 2), Auth system (Dev 3)
- Blocks: Testing (Dev 5), Frontend integration (Dev 4)

2.2 Developer 2 — Backend and Data Engineer

Responsibilities:

- Database schema design
- Models and migrations
- Vector database and RAG pipeline

Main Tasks:

- Design HR database schema
- Implement models
- Set up migrations
- Build document ingestion pipeline
- Implement HR policy search

Dependencies:

- Independent at start
- Blocks: Dev 1, Dev 5

2.3 Developer 3 — Security, Auth and API Engineer

Responsibilities:

- Authentication (JWT, OAuth2)
- Authorization (RBAC)
- API gateway middleware
- Rate limiting and logging

Main Tasks:

- Implement login and token system
- Build JWT middleware
- Define roles and permissions
- Enforce authorization rules
- Add rate limiting

Dependencies:

- Independent at start
- Blocks: Dev 1, Dev 4

2.4 Developer 4 — Frontend Engineer

Responsibilities:

- Chat interface
- API integration
- User experience

Main Tasks:

- Build chat UI
- Implement authentication UI
- Add conversation history
- Display tool activity
- Ensure responsive design

Dependencies:

- Requires: Dev 1 and Dev 3
- Can start with mock APIs

2.5 Developer 5 — Memory, Testing and DevOps Engineer

Responsibilities:

- Memory system
- Testing
- Deployment and CI/CD
- Monitoring

Main Tasks:

- Implement memory manager
- Build summarization system
- Write tests
- Configure Docker deployment
- Set up CI/CD

Dependencies:

- Requires: Dev 1 and Dev 2

3 Task Dependencies Overview

```
Dev 2 (Database) ----\
                      --> Dev 1 (Orchestrator) --> Dev 5 (Testing)
Dev 3 (Auth) --------/

Dev 1 + Dev 3 ------> Dev 4 (Frontend)
```

4 Parallel Development Strategy

Weeks 1–2:

- Dev 2: Database and RAG pipeline
- Dev 3: Authentication system
- Dev 5: Docker and CI/CD
- Dev 4: UI design
- Dev 1: LLM experimentation

Weeks 3–5:

- Dev 1: Orchestrator integration
- Dev 5: Memory system
- Dev 4: API integration

Weeks 6+:

- Dev 5: Testing and security
- All: Integration and debugging

5 Conclusion

This distribution ensures parallel development, clear ownership, minimal blocking, and a production-ready system.